

Positive non-atomic minimal extrapolations are never unique

Full answer to the explicit uniqueness questions in arXiv:1601.05761

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Status. Candidate full solution, likely valid, pending human review. The two explicit uniqueness questions in Examples 3.8 and 3.9 of the source are both answered negatively. The general theorem also settles every positive non-atomic case when the sampled frequency set is finite and contains zero.

1 Source questions

Benedetto and Li minimize total variation among measures having prescribed Fourier coefficients on a finite set $\Lambda \subset \mathbb{Z}^d$ [1]. On source PDF page 20, Example 3.8 says that their theory cannot determine whether a positive continuous singular Cantor measure σ_q is the unique minimal extrapolation. Example 3.9 asks the same question for $\mu = \sigma_A + \sigma_B$, where σ_A and σ_B are Haar surface measure on the horizontal circles $x_2 = 0$ and $x_2 = 1/2$ in $\mathbb{T}^2$.

$$\sigma_q(m) = (-1)^{m_2} \prod_{k=1}^{\infty} \cos(\pi m q^{-k} (1-q)),$$

see [Zyg59, pages 195-196]. In particular, for any integer $n \geq 1$, we have

$$\widehat{\sigma}_q(q^n) = (-1)^{q^n} \prod_{k=1}^{\infty} \cos(\pi q^{-k} (1-q)),$$

which is convergent and independent of n . Since $\widehat{\sigma}_q(0) = \|\sigma_q\| = 1$, we immediately see that $\varepsilon = 1$ and $\sigma_q \in \mathcal{E}$. Again, we cannot determine whether σ_q is the unique minimal extrapolation because Theorem 1.4 cannot handle the case $\#\Gamma = 1$, see Remark 2.10.

Example 3.9. Let $\sigma_A, \sigma_B \in M(\mathbb{T}^d)$ be the surface measures of the Borel sets $A = \{x \in \mathbb{T}^2: x_2 = 0\}$ and $B = \{x \in \mathbb{T}^2: x_2 = 1/2\}$, respectively. Let $\mu = \sigma_A + \sigma_B$, and $\Lambda = \{-2, -1, \dots, 1\}^2$. Then,

$$\widehat{\mu}(m) = \int_0^1 e^{2\pi i m_1 t} dt + \int_0^1 e^{2\pi i (m_1 t + m_2/2)} dt = \delta_{m_1, 0} + (-1)^{m_2} \delta_{m_1, 0}.$$

We immediately see that $\varepsilon = \|\widehat{\mu}\|_{\ell^\infty(\Lambda)} = \|\mu\| = 2$, which implies $\mu \in \mathcal{E}$. Then, $\Gamma = \{(0, 0), (0, 2), (0, -2)\}$, and, by Theorem 1.4b, the minimal extrapolations are supported in $\{x \in \mathbb{T}^2: x_2 = 0\} \cup \{x \in \mathbb{T}^2: x_2 = 1/2\}$. Determining whether μ is the unique minimal extrapolation is beyond the theory we have developed herein, and we shall examine this uniqueness problem in [BL16].

There is a minor internal inconsistency in Example 3.9: the printed set is $\{-2, -1, \dots, 1\}^2$, while the displayed set Γ contains frequencies with second coordinate 2. The explicit answer below matches the data on the larger rectangle $\{-2, -1, 0, 1, 2\}^2$, and therefore covers both readings.

2 Proof intuition

The zero Fourier coefficient is decisive. For a positive measure μ , it equals $\|\mu\|$, so every extrapolation has norm at least $\|\mu\|$. Thus every positive measure sharing the sampled moments is automatically another minimal extrapolation. Finitely many Fourier constraints impose only finitely many real linear equations on a bounded density perturbation h . A non-atomic measure can be split into more positive-measure pieces than there are equations, leaving a nonzero kernel vector. After scaling h , both $(1+h)\mu$ and $(1-h)\mu$ remain positive and share all sampled Fourier data.

For the surface-measure example this abstract perturbation can be made fully explicit: replace uniform measure in the x_1 direction by the uniform five-point rule on the fifth roots of unity. It integrates every Fourier mode $m_1 = -2, -1, 0, 1, 2$ exactly.

3 General non-uniqueness theorem

For a finite positive measure μ on $\mathbb{T}^d$, write

$$\widehat{\mu}(m) = \int_{\mathbb{T}^d} e^{-2\pi i m \cdot x} d\mu(x).$$

Theorem 1. *Let μ be a nonzero positive non-atomic finite measure on $\mathbb{T}^d$, and let $\Lambda \subset \mathbb{Z}^d$ be finite with $0 \in \Lambda$. Then μ is a minimal extrapolation of $\widehat{\mu}|_{\Lambda}$, but it is not the unique minimal extrapolation. In fact, there are distinct positive measures μ_+ and μ_- such that*

$$\widehat{\mu_+}(m) = \widehat{\mu_-}(m) = \widehat{\mu}(m) \quad (m \in \Lambda), \quad \|\mu_+\| = \|\mu_-\| = \|\mu\|.$$

Proof. Let $r = 2|\Lambda| + 1$. Non-atomicity permits a measurable partition $\mathbb{T}^d = E_1 \sqcup \dots \sqcup E_r$ modulo μ -null sets, with $\mu(E_j) > 0$ for every j . For $c = (c_1, \dots, c_r) \in \mathbb{R}^r$, put $h_c = \sum_j c_j \mathbf{1}_{E_j}$. The conditions

$$\int_{\mathbb{T}^d} e^{-2\pi i m \cdot x} h_c(x) d\mu(x) = 0 \quad (m \in \Lambda) \tag{1}$$

form at most $2|\Lambda|$ homogeneous real linear equations in the r real unknowns c_j . Hence they admit a nonzero solution c . Rescale it so that $\|h_c\|_{\infty} \leq 1/2$, and write $h = h_c$.

Define $d\mu_+ = (1+h)d\mu$ and $d\mu_- = (1-h)d\mu$. These measures are positive and distinct, and (1) gives their asserted Fourier data. Since $0 \in \Lambda$, every extrapolation ν satisfies

$$\|\nu\| \geq |\widehat{\nu}(0)| = |\widehat{\mu}(0)| = \mu(\mathbb{T}^d) = \|\mu\|.$$

On the other hand, positivity and the zero-frequency equality give $\|\mu_{\pm}\| = \mu_{\pm}(\mathbb{T}^d) = \mu(\mathbb{T}^d)$. Thus μ , μ_+ , and μ_- are all minimal extrapolations, proving non-uniqueness. $\square$

Corollary 2 (Answer to Example 3.8). *Under the standing finite-frequency assumption of the source, its positive continuous singular Cantor measure σ_q is not the unique minimal extrapolation from any finite Λ containing 0.*

Proof. The source constructs σ_q as a positive continuous measure of mass one. It is therefore non-atomic, and Theorem 1 applies. $\square$

4 Explicit answer to the surface-measure example

Let σ_A and σ_B be normalized Haar measure on $A = \{x_2 = 0\}$ and $B = \{x_2 = 1/2\}$, and set $\mu = \sigma_A + \sigma_B$ as in Example 3.9. Define the positive atomic measure

$$\nu = \frac{1}{5} \sum_{j=0}^4 (\delta_{(j/5, 0)} + \delta_{(j/5, 1/2)}).$$

Corollary 3 (Answer to Example 3.9). *The source measure μ is not the unique minimal extrapolation. The measure ν is a distinct minimal extrapolation for the printed frequency set and also for the symmetric set $\{-2, -1, 0, 1, 2\}^2$ suggested by the displayed Γ .*

Proof. For $m = (m_1, m_2) \in \mathbb{Z}^2$,

$$\widehat{\mu}(m) = \mathbf{1}_{\{m_1=0\}}(1 + (-1)^{m_2}),$$

whereas

$$\widehat{\nu}(m) = \left(\frac{1}{5} \sum_{j=0}^4 e^{-2\pi i m_1 j/5} \right) (1 + (-1)^{m_2}).$$

The parenthesized roots-of-unity sum is one when 5 divides m_1 and zero otherwise. Hence $\widehat{\nu}(m) = \widehat{\mu}(m)$ whenever $|m_1| \leq 2$, with no restriction on m_2 . In particular the equality holds on either reading of the source frequency set.

Both measures are positive with total mass two, while every extrapolation has norm at least the absolute value 2 of its zero Fourier coefficient. Thus both are minimal. Finally, ν is atomic and μ is non-atomic, so they are distinct. $\square$

5 Verification, novelty, and scope

The accompanying script `code/verify_five_point_extrapolation.py` evaluates the explicit Fourier formulas on $m_1 = -2, \dots, 2$ and a larger range of m_2 ; the exact roots-of-unity proof above is independent of the computation.

A bounded novelty search covered the run indexes, exact phrases from Examples 3.8–3.9, the promised title “Super-resolution of singular continuous measures on $\mathbb{T}^d$ ”, and close arXiv searches for positive non-atomic minimal extrapolations and finite Fourier moments. The promised sequel remains listed as “in preparation” in the later journal version and a 2018 dissertation; no primary source giving these answers was found. The underlying finite-moment argument is elementary, so novelty confidence is moderate even though correctness confidence is high.

The result fully settles the two explicit source uniqueness questions. The general theorem requires positivity, non-atomicity, a finite frequency set, and inclusion of zero; it does not classify signed or complex singular continuous measures. Human review should verify the real dimension count in (1) and the scope attached to the word “full”.

References

- [1] J. J. Benedetto and W. Li, *Super-resolution by means of Beurling minimal extrapolation*, arXiv:1601.05761; Appl. Comput. Harmon. Anal. 48 (2020), 117–148.